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**FPT UNIVERSITY
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AUTOMATIC FIRE ALARM AND EXTINGUISHING SYSTEM

Final Project Report

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Automatic Fire Alarm and Extinguishing System

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Abstract

This paper presents an intelligent IoT-based fire alarm system designed to enhance safety in residential and industrial environments. The system integrates a multi-sensor array—comprising flame (KY-026), smoke (MP-2), and temperature (DHT22) sensors—leveraging a dual-axis active scanning mechanism powered by SG-90 micro servos to eliminate detection blind spots. By employing a multi-condition logic, the system ensures high detection accuracy while significantly minimizing false alarms. The architecture features a load-sharing design between an Arduino for real-time hardware execution and a NodeMCU ESP8266 for stable IoT connectivity. Experimental results demonstrate that the system effectively identifies fire hazards and activates protective measures (buzzer and water pump) within 5 seconds, providing a reliable and cost-effective solution for early fire prevention and remote monitoring.

Index Terms

Internet of Things (IoT), Fire Alarm System, Multi-sensor Fusion, NodeMCU ESP8266, Embedded Systems, Automated Fire Suppression.

I. INTRODUCTION

In the first two months of 2026, 512 fires occurred nationwide, with 36% of them originating in residential buildings. January 2026 recorded 97 incidents, followed by 89 in February. Objectively, the most frequent cause of fires is negligence in the maintenance and use of electrical equipment. Subjectively, a lack of precautionary habits leads to hazardous situations, such as unattended overnight charging of devices or auditory obstructions from personal entertainment, which prevent occupants from noticing early signs of electrical malfunctions. Current residential fire alarms typically rely on isolated smoke or temperature sensors. Due to their limited precision, false alarms are frequent, making the integration of automatic

extinguishing systems or direct emergency service calls high-risk and impractical for domestic use.

Recognizing these limitations, this project develops an Automatic Fire Warning and Solution based on Internet of Things (IoT) technology. The primary objective is to create a system that detects fire hazards with high precision through multi-sensor fusion while initiating an immediate automated first response. By combining a relay-controlled water pump for local suppression and a high-decibel buzzer for alerting, the system provides a comprehensive safety net.

This report details the hardware configuration, the software logic, and the experimental results of the smart fire alarm system. Through this application of IoT, we aim to provide an efficient, low-cost, and high-response solution to mitigate the devastating effects of fire disasters in modern living environments.

Key Features

- **Multi-Sensor Fusion & Logic:** The system integrates three types of environmental data: Infrared signals from Flame Sensors (KY-026), gas concentration from the MP-2 Smoke Sensor, and thermal data from the DHT22 Temperature Sensor. A specialized algorithm ($\text{activeCount} \geq 2$) ensures that the alarm only triggers when at least two hazardous conditions are confirmed, effectively minimizing false alarms.
- **Dual-Axis Active Scanning:** Utilizing two SG-90 Servo motors, the sensor array performs a dynamic horizontal and vertical sweep. This active monitoring mechanism significantly expands the detection range and eliminates "blind spots" common in static fire detectors.
- **Dual-Controller Stability:** The architecture employs a load-sharing design between an Arduino (for real-time hardware control) and a NodeMCU ESP8266 (dedicated to IoT communication). This ensures the system remains responsive even during high-traffic data transmission.
- **Automated First Response:** Upon fire confirmation, the system integrates a Water Pump (via Relay) for immediate localized fire suppression, while a high-decibel Buzzer ensures occupants are alerted instantly.
- **Real-time IoT Monitoring:** Through the ESP8266 module, the system provides 24/7 remote surveillance. Users receive instant notifications on their mobile devices, allowing for immediate intervention.

II. METHODOLOGY

A. Research Design and Approach

The project employs an engineering design combined with experimental evaluation. The process is divided into four main phases:

- **System Design:** Defining the overall architecture and data flow.
- **Hardware Integration:** Selection and physical connection of sensors and actuators.
- **Software Development:** Programming the control logic and cloud communication.
- **Testing and Validation:** Evaluating system latency and detection accuracy.

B. System Models and Block Diagram

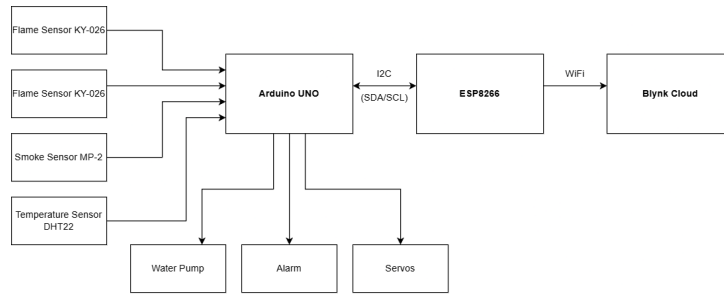


Fig. 1. Block diagram of the developed system.

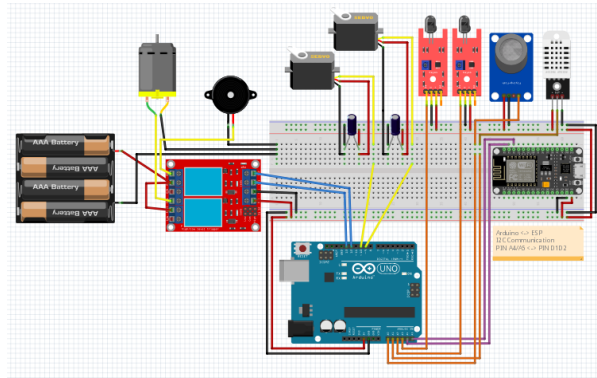


Fig. 2. System prototype layout.

C. Hardware Design and Connections

This section describes the hardware interconnection between the Arduino Uno, NodeMCU ESP8266, and various sensors/actuators.

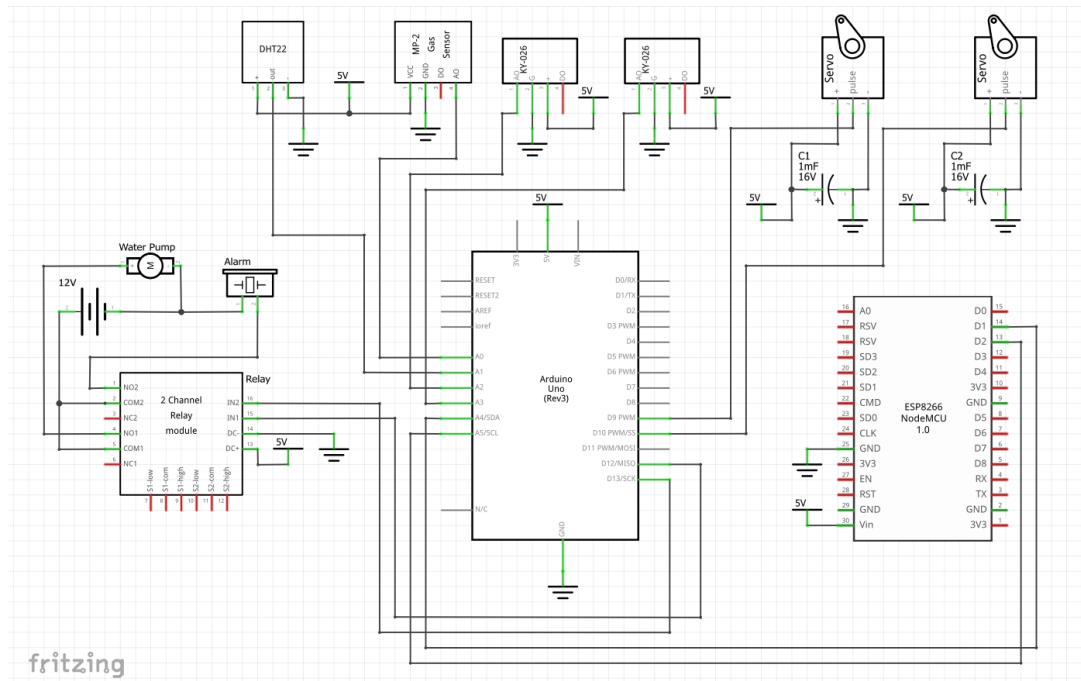


Fig. 3. System schematic diagram featuring Arduino Uno and ESP8266.

The specific pin assignments for each component are detailed in Fig 4 . This setup ensures stable communication and power distribution across the system.

Component	Arduino Pin	NodeMCU Pin	Notes
Flame Sensor KY-026	Pin A2	-	Analog input (0-1023).
Flame Sensor KY-026	Pin A3	-	Analog input (0-1023).
Smoke Sensor MP-2	Pin A0	-	Analog input (0-1023).
Temp Sensor DHT22	Pin A1	-	Analog input (0-1023).
Servo Motor SG-90	Pin 9	-	PWM signal for scanning rotation.
Servo Motor SG-90	Pin 10	-	PWM signal for scanning rotation.
Water Pump	Pin 12	-	Connect via Relay.
Buzzer	Pin 13	-	Connect via Relay.
SDA (Data)	Pin A4	D1 (GPIO 5)	
SCL (Clock)	Pin A5	D2 (GPIO 4)	
Common Ground	GND	GND	Crucial: Both boards must share GND.

Fig. 4. Pin Mapping between Arduino Uno, NodeMCU, and Components.

D. Software Programming

[Click here to view code](#)

E. Programming Flowchart

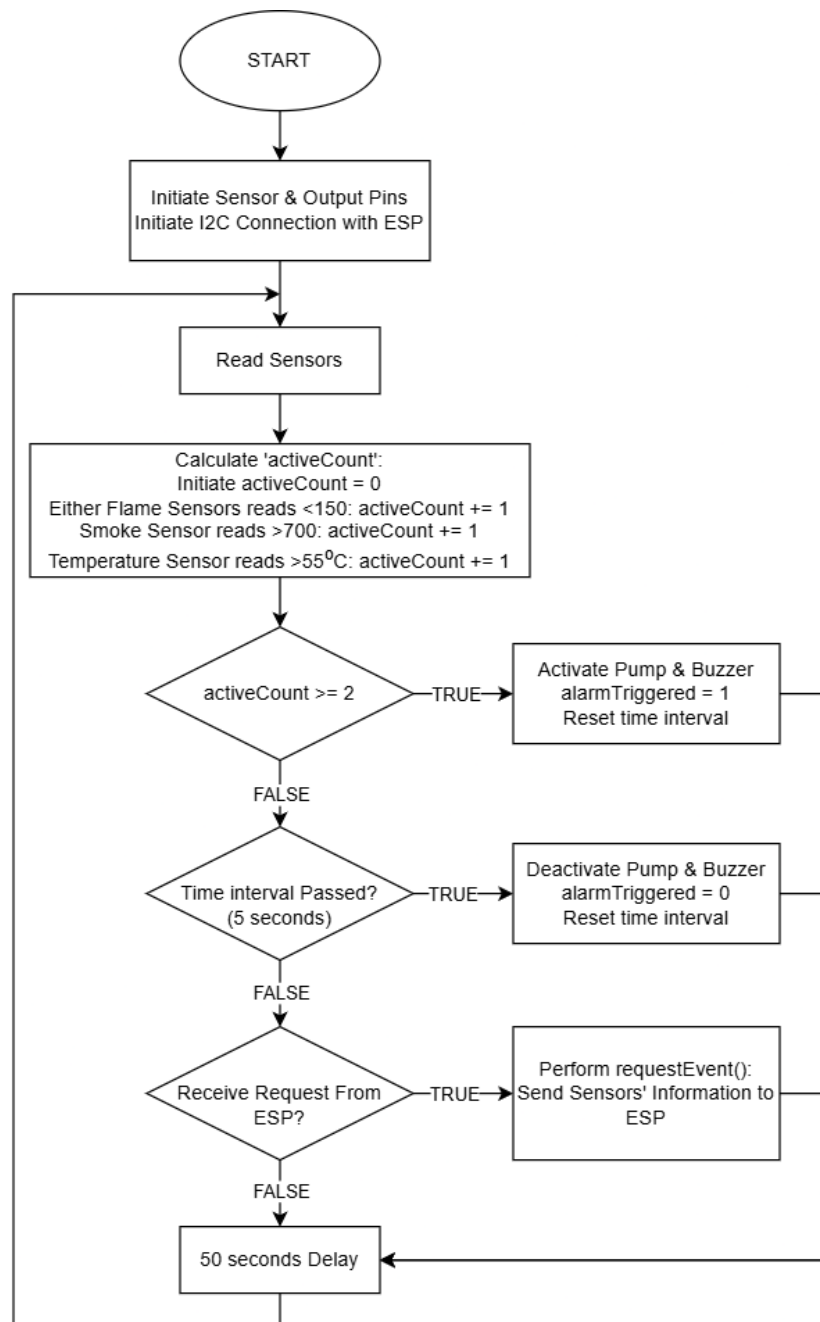


Fig. 5. Flowchart of Arduino

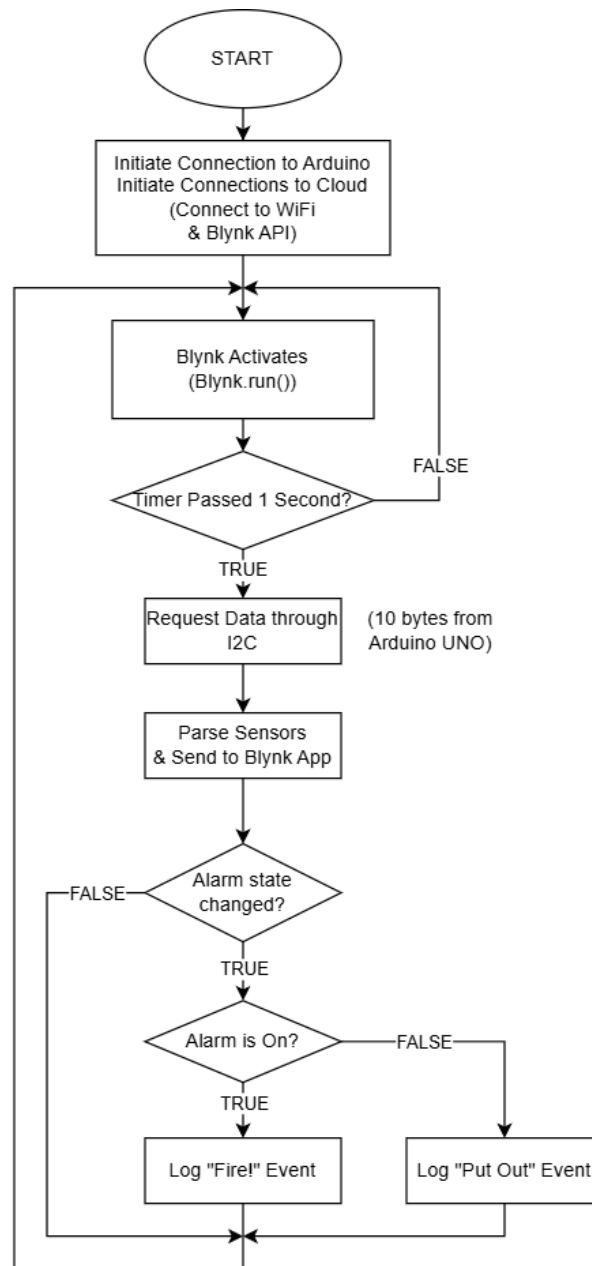


Fig. 6. Flowchart of ESP8266

F. System Procedure

The operational sequence of the system is summarized as follows:

- **System Initialization:** Establish I^2C communication between NodeMCU (Master) and Arduino Uno (Slave), and initialize the DHT22 single-wire digital bus.
- **Scanning & Monitoring:** A servo-actuated mechanism continuously sweeps the KY-026 sensors to maximize spatial coverage.
- **Data Processing:** Arduino evaluates telemetry against three thresholds: flame (Analog < 150), smoke density (Analog > 700), and thermal safety limits.

- **Sensor Fusion:** A multi-sensor fusion algorithm validates an active fire state only when at least two sensor criteria are simultaneously satisfied to prevent false positives.
- **Local Actuation:** Upon validation, the Arduino triggers a two-channel relay to activate the 12V submersible pump and acoustic buzzer for immediate suppression.
- **Cloud Telemetry:** NodeMCU uses a software Ticker interrupt to poll data and transmit real-time alerts to the Blynk IoT server via TCP/IP stack.
- **Hysteresis & Reset:** A five-second temporal hysteresis window ensures the anomaly is neutralized before deactivating the relay and logging a "System Secured" status.

III. IMPLEMENTATION AND RESULTS

A. *Prototype Implementation*

Click [here](#) to watch video demo

B. *Experimental Setup and Testing*

To evaluate the reliability of the system, two primary fire scenarios were simulated in a controlled environment. The focus was on testing the scanning capability of the Servo motor and the threshold calibration of the sensors.

1) *Test Scenarios:*

- **Center-Room Fire:** The fire source was placed directly in front of the system's initial position. This test confirmed that the sensors could trigger an immediate response when the threat was within the direct line of sight.
- **Corner-Room Fire:** The fire source was placed at a 45-degree angle. This scenario successfully validated the **Servo Motor's scanning capability**, proving that the system can locate a fire even when it is not initially centered.

2) *Distinguishing Daily Activities vs. Actual Fire:* One of the critical objectives was to ensure the system could differentiate between non-hazardous heat sources and actual fire outbreaks:

- **Normal Daily Activity (Low-level flame):** During testing with small, controlled flames (e.g., a cigarette lighter or a small candle at a distance), the system monitored the values but did not trigger the alarm. This proves the sensitivity is well-calibrated to avoid false alarms.
- **Actual Fire Outbreak:** When the flame intensity increased to a hazardous level, the integrated logic (combining Flame, Heat, and Smoke data) reached the pre-defined thresholds, confirming a valid emergency.

3) **System Response Sequence:** Once a fire was confirmed, the following response sequence was recorded:

- 1) **Local Alert:** The Buzzer was immediately activated with a high-frequency siren.
- 2) **Suppression:** The Relay module switched on, powering the Water Pump to initiate early-stage suppression.
- 3) **Remote Notification:** Simultaneously, a Push Notification (Pop-up) was sent to the user's smartphone via the Blynk IoT platform, ensuring real-time awareness regardless of the user's location.

IV. DISCUSSION

A. *Limitations of the System*

Despite the successful demonstration of the prototype, several technical and operational limitations were identified during the testing phase:

- **Resource Dependency and Lack of Redundancy:** The system currently lacks substitute options for critical resources. In a real-world fire scenario, power outages are frequent; however, the device does not have a secondary power source (e.g., a backup battery). Furthermore, the fire suppression mechanism is highly vulnerable to water shortages or low pressure in the main supply.
- **High Dependency on Sensor Availability** ($activeCount \geq 2$): According to the system's flowchart logic, the alarm and pump are only triggered when at least two sensors detect an anomaly simultaneously. This creates a significant risk: if one or two sensors (Flame, Smoke, or Temperature) malfunction or disconnect, the system may fail to reach the required threshold to activate the emergency response, even if a fire is present.
- **Simple Risk Analysis and Potential for Errors:** The system relies on fixed static thresholds for its logic. While it can distinguish between fire and normal conditions, this simplicity may lead to measurement errors.
- **Complex Installation and Scalability:** As a prototype, the current setup involves a complicated installation process with many interconnected components. This makes it difficult to scale the system for larger buildings or to maintain and replace individual parts without disrupting the entire network.
- **Hardware Stability and Unstable Wiring:** The use of jumper wires and breadboards leads to an unstable wiring system. These connections are susceptible to signal noise

and physical disconnection, especially under the high-heat conditions of a fire. This structural fragility limits the reliability of the device in industrial or high-risk domestic applications.

B. Future Improvements

To transform the prototype into a more practical and user-centric fire safety solution, the following enhancements are proposed:

- **Remote Manual Override:** Implementing a two-way communication feature that allows users to intervene remotely. This includes a "Force Stop" function to silence false alarms or a "Manual Trigger" to activate the water pump via a mobile app if a fire is visually confirmed through external cameras.
- **Local Data Logging (Black Box):** Incorporating an **SD Card module** to act as an offline "black box." This would ensure that environmental data is continuously recorded even during a Wi-Fi failure, providing critical information for post-incident investigations and forensic analysis.

V. CONCLUSION

The IoT-based weather station successfully monitors critical environmental parameters. Future work will focus on integrating solar power for better energy efficiency and adding more gas sensors (MQ series) for comprehensive air quality analysis.

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